20 A steel wire has a length of 300 cm and a cross-sectional area of 0.50 mm^2. The Young modulus of steel is 2.0×10^{11} Pa.

One end of the wire is attached to a fixed point. A load of 10 N is hung from the other end. The wire obeys Hooke's law.

What is the extension of the wire?

A 3.0×10^{-7} m

B 3.0×10^{-5} m

C 3.0×10^{-4} m

D 3.0×10^{-2} m